

Supplementary Material

SPLITGUARD-AD: Executable Leakage Auditing for Slice-Based Alzheimer’s Disease MRI Deep Learning Benchmarks

S1 Consolidated training hyperparameters

Table S1 lists the full per-cohort training configuration referenced in the main text.

Table S1: Consolidated training hyperparameters across the three tiers and both backbones.

Parameter	Kaggle	OASIS-1	ADNI1
Backbones	ResNet-18; DenseNet-121	ResNet-18; DenseNet-121	ResNet-18 DenseNet-121 primary; architecture arm
Initialisation	ImageNet	ImageNet	ImageNet
Input	$3 \times 224 \times 224$	$3 \times 224 \times 224$	$3 \times 224 \times 224$
Normalisation	ImageNet mean/std	ImageNet mean/std	ImageNet mean/std
Augmentation	flip + $\pm 7^\circ$	flip + $\pm 7^\circ$	flip + $\pm 7^\circ$
Epochs	30	20	15 (all arms)
Optimiser	AdamW	AdamW	AdamW
Learning rate	3×10^{-4}	3×10^{-4}	10^{-4}
LR schedule	Cosine anneal	Cosine anneal	Cosine anneal
Weight decay	10^{-2}	10^{-2}	10^{-2}
Batch size	32	32	16
Loss	inverse-freq BCE	inverse-freq BCE	inverse-freq BCE
Seed array	{0, 1, 2, 3, 42}	{0, 1, 2, 3, 42}	{0, 1, 2, 3, 4}
Backend (every arm)	CUDA H100	CUDA H100	CUDA H100
Mixed precision	AMP	off	off

Every arm ran in one code state on one GPU. Mixed precision is the one setting that differs by cohort: the Tier-1 trainer enables automatic mixed precision on CUDA and the OASIS-1 and ADNI trainers do not. Within any one cohort every protocol is trained under identical settings, which is the condition the inflation-gap comparison requires. The volumetric ADNI arm uses a 3D ResNet-18 on 128^3 skull-stripped, z-scored volumes, batch size 4, AdamW at 10^{-4} with weight decay 10^{-2} , 15 epochs.

S2 Operating-point sensitivity table (ADNI)

S3 Mapping SplitGuard-AD outputs to standard reporting domains

Table S3 maps SPLITGUARD-AD’s pre-training audit, model audit, and reporting artefacts to the standard reporting domains covered by TRIPOD+AI and PROBAST domain 4 (Analysis). We

Table S2: Operating-point sensitivity table for the ADNI converter-inclusive arm. **Every threshold is selected on the validation partition and applied to test**; no operating point is chosen by scanning the test ROC. The fixed-rate rows are mean $\pm$ SD across five seeds and the Youden rows are seed means, with Δ the Protocol A minus Protocol C difference.

Operating point	Random	Subject-only	Component-safe	Δ
<i>Sensitivity at fixed specificity</i>				
Spec = 0.80	0.919 $\pm$ 0.028	0.717 $\pm$ 0.122	0.649 $\pm$ 0.084	+0.270
Spec = 0.85	0.901 $\pm$ 0.024	0.640 $\pm$ 0.109	0.616 $\pm$ 0.100	+0.285
Spec = 0.90	0.866 $\pm$ 0.046	0.551 $\pm$ 0.105	0.535 $\pm$ 0.103	+0.331
Spec = 0.95	0.810 $\pm$ 0.057	0.382 $\pm$ 0.055	0.386 $\pm$ 0.099	+0.424
<i>Specificity at fixed sensitivity</i>				
Sens = 0.85	0.948 $\pm$ 0.028	0.567 $\pm$ 0.114	0.563 $\pm$ 0.108	+0.385
Sens = 0.90	0.898 $\pm$ 0.061	0.439 $\pm$ 0.124	0.488 $\pm$ 0.141	+0.410
Sens = 0.95	0.782 $\pm$ 0.125	0.299 $\pm$ 0.111	0.403 $\pm$ 0.162	+0.379
<i>Youden-optimal threshold (selected on validation)</i>				
J	0.785	0.458	0.533	+0.252
Sens at J	0.836	0.768	0.682	+0.153
Spec at J	0.950	0.690	0.851	+0.099

deliberately list domains rather than guideline item numbers because TRIPOD+AI numbering is sensitive to the model-development vs. model-validation arm of the guideline; the substantive mapping stays valid regardless of which arm is being checked. SPLITGUARD-AD is not a replacement for the broader guidelines; it operationalises the subset concerning data partitioning, reproducibility, calibration, and discrimination as machine-checkable outputs.

Domains not directly operationalised by SPLITGUARD-AD (title/abstract, background and objectives, outcome/predictor variable definition, model specification) remain author- and reviewer-side checklist work; SPLITGUARD-AD complements rather than substitutes for the broader guideline. PROCAST domain 4 (Analysis) risk-of-bias items concerning sample handling, missing data, and overoptimism in performance estimation are addressed by the same pre-training audit outputs above.

The same outputs answer the CLAIM [1, 2] items a leakage audit touches, listed here by topic rather than by number because the 2024 revision renumbered them. *Data sources and provenance*: the per-image manifest records where each image came from and which identifiers were available, and for Tier 1 the provenance is recovered by content matching rather than assumed (§2.3.1 of the main text). *Data partitioning*: the split rule is frozen in a versioned manifest, the audit reports participant, session and component overlap for each partition, and the compositional check reports how the partitions differ where they are disjoint. *Reference standard*: the label join is audited, with the share of labels resting on a date-proximity match reported alongside the exact-key share. *Evaluation*: performance is reported per protocol with both image-level and participant-level aggregation, calibration, and sensitivity at a fixed specificity. *Failure analysis and reproducibility*: the identity probe, the provenance-degradation sweep and the released rebuild pipeline cover the failure-mode and reproducibility items.

S4 Dose-response mixed-effects model specification

The dose-response OLS slope in the main text (Results: leakage as a dose) is a descriptive calibration constant. To obtain an inferential quantity that respects the paired structure of the 25

Table S3: SPLITGUARD-AD outputs mapped to standard reporting domains.

Reporting domain	SplitGuard-AD output	Main-text section
Source of data & provenance	Per-image manifest with subject/session/series IDs and preprocessing lineage; leakage-graph edges over subject/source-volume/near-duplicate.	Methods: cohorts
Eligibility & splitting rules	Component-safe split rule with class-stratification target; orphan-component exclusions audited.	Methods: audit instrument
Sample-size handling & inference	Five-seed array with “descriptive of seed-level variability” labelling and hierarchical (subject $\times$ seed) bootstrap as the inferential cross-check.	Results: statistical conventions
Missing-data audit	Pre-training audit flags unparsable filename keys (Tier 1), missing CDR (Tier 2), and unresolved diagnosis (Tier 3).	Methods: cohorts
Statistical methods	Paired-seed bootstrap with direction-preservation count; hierarchical bootstrap as the defensible inferential quantity.	Results: statistical conventions
Calibration	Brier score and expected calibration error reported across protocols; monotone degradation with leakage protection.	Results: ADNI robustness
Discrimination	AUROC (image- and subject-level), sensitivity at fixed specificity, ROC curves with operating-point overlay.	Results: ADNI arm and screening operating point
Subgroup performance	Sex-stratified AUROC with paired-seed bootstrap (exploratory).	Supplementary S6, Figure S1
Clinical-utility translation	Sensitivity at fixed specificity = 0.90; predictive values at two prevalence anchors.	Results: screening operating point
Reproducibility & versioning	Frozen split manifests (seed-byte-reproducible CSV), versioned audit report, GitHub release.	Data and Code Availability
Limitations & bias declaration	Single-cohort headline scope, $n = 5$ seeds, 2D-slice representation, component-marginal CI straddles zero, subgroup small-sample.	Discussion: Limitations
Code & artefact availability	Apache-2.0 release of manifest builders, leakage-graph construction, split generation, audit, probe, and bootstrap scripts.	Data and Code Availability

(seed, overlap) observations, we additionally fit the random-intercept model

$$\text{AUROC}_{ij} = \beta_0 + \beta_1 p_{ij} + u_i + \varepsilon_{ij}, \quad u_i \sim N(0, \sigma_u^2), \quad \varepsilon_{ij} \sim N(0, \sigma^2),$$

by iterated feasible GLS under compound symmetry (`analyze_dose_response_mixed_effects.py` in the release repository). The fitted variance ratio is $\hat{\rho} = \hat{\sigma}_u^2 / \hat{\sigma}^2 = 0.649$ on ResNet-18, confirming non-negligible between-seed variation.

Because the design is *balanced* (every seed contributes exactly one observation at each of the five overlap levels) the GLS and OLS point estimates coincide by construction: $\hat{\beta}_1 = 0.1065$ under both. The random-intercept structure therefore does not move the slope here; what it changes is the uncertainty attached to it, and that is where the inferential content lies.

Inference uses a cluster-robust sandwich standard error over seed with the finite-cluster scaling $\frac{G}{G-1} \cdot \frac{N-1}{N-k}$ and a $t(G-1)$ critical value. With $G = 5$ seeds the correct multiplier is $t_{0.975}(4) = 2.776$, not the normal 1.960; using the normal quantile at this cluster count understates the interval by roughly 40%. The resulting estimates are: ResNet-18 $\hat{\beta}_1 = 0.1065$, cluster-robust SE 0.0107, 95% CI

[0.0768, 0.1362], with a non-parametric seed-cluster bootstrap ($B=10,000$) giving the comparable interval [0.0886, 0.1249]; DenseNet-121 $\hat{\beta}_1 = 0.0655$, SE 0.0161, 95% CI [0.0209, 0.1102]. Both intervals exclude zero, and they overlap substantially, so the two architectures’ dose-response slopes are not distinguishable at five seeds.

Spearman ρ on the pooled 25-point (p , AUROC) pairs is 0.875 for ResNet-18 and 0.553 for DenseNet-121; per-seed AUROC is monotonically non-decreasing in overlap in 4 of 5 ResNet-18 seeds and 0 of 5 DenseNet-121 seeds, consistent with the smaller ResNet-vs-DenseNet inflation-gap asymmetry reported in the ADNI arm.

S5 Metadata linkage-integrity artefact

The full scan-to-visit linkage accounting for the ADNI headline cohort is released as a per-scan CSV (`adni_linkage_audit.csv`, one row per scan), produced by `scripts/build_linkage_audit.py`. Columns:

- `scan_uid`: image identifier (not participant-derived);
- `ptid_hash`: PBKDF2-HMAC-SHA256 of the PTID under a secret per-release salt (600,000 iterations). The salt is never written to the released artefacts;
- `acq_date`: acquisition date, retained because the join cannot be audited without it;
- `join_tier`: `exact_viscode` (direct (RID, VISCODE) match), `date_proximity` (visit key failed; label taken from the nearest clinical visit within the 180-day tolerance), or `unresolved`;
- `label_source`: the ADNI column the diagnosis came from;
- `diagnosis_group`: resolved CN / MCI / AD / unknown;
- `in_cnad_universe`: whether the scan enters the CN-vs-AD task.

Tier counts on the released cohort: `exact_viscode` 1,695 (77.7%), `date_proximity` 485 (22.2%), `unresolved` 2 (0.1%). The companion summary (`reports/audits/adni/adni_linkage_audit_summary.json`) records the same counts plus the KDF parameters.

We do not record a candidate-visit count or a minimum date-difference per scan: the manifest builder retains only the tier it resolved to, so those quantities are not recoverable without re-running the join with extra instrumentation. We state this rather than describing fields the released file does not contain.

S6 Sex-stratified performance on ADNI (exploratory)

Stratifying test-set AUROC by sex within each protocol (Figure S1) surfaces a female-minus-male AUROC delta of +0.102 [+0.068, +0.136] under subject-only and +0.115 [+0.046, +0.181] under component-safe SPLITGUARD-AD that collapses to -0.020 [-0.055, +0.015] under the leaky protocol, where the sign holds in only 8,893/10,000 resamples. We report the two deltas descriptively and do not test their difference. The gap does not appear to be explained by test-set class imbalance: the female AD share is lower than the male share under all three protocols. Sex differences in AD atrophy trajectories and cognitive reserve are documented in the wider literature [3, 4, 5], but per-cell sample sizes here are small (~ 8 –13 subjects per sex $\times$ age cell) and no adjustment for APOE, education, site, or scanner is performed; **this finding is exploratory and hypothesis-generating only**, and it should not be interpreted as evidence of a biological sex effect without

external-cohort replication. The relevant point is methodological rather than biological: leaky evaluation would prevent detection of any such effect if one existed.

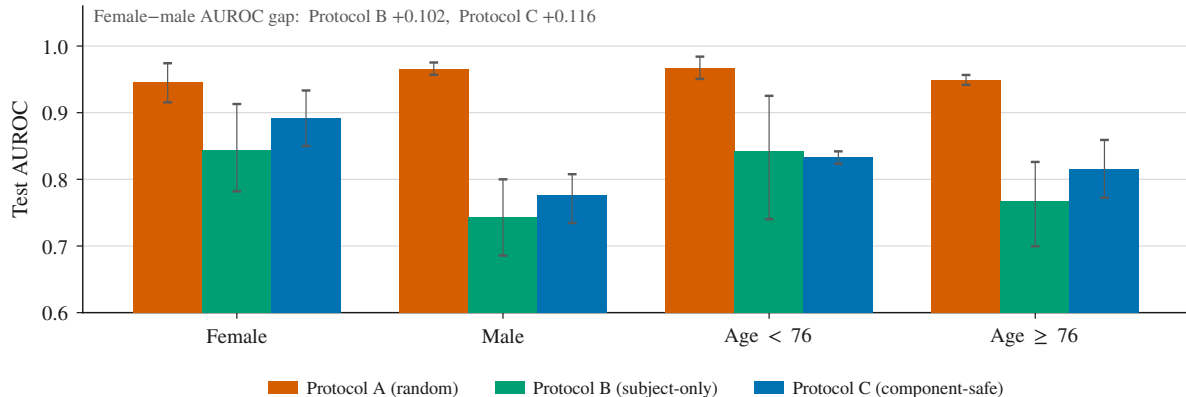


Figure S1: ADNI per-subgroup test AUROC across five seeds (paired-seed bootstrap intervals, $B = 10,000$; descriptive). Both honest protocols show a $\sim +0.11$ female-male AUROC gap; under the leaky protocol the same contrast has a point estimate near zero. Reported as exploratory, pending external-cohort replication.

References

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- [3] E. E. Sundermann, P. M. Maki, L. H. Rubin, R. B. Lipton, S. Landau, and A. Biegon. Female advantage in verbal memory: Evidence of sex-specific cognitive reserve. *Neurology*, 87(18):1916–1924, 2016.
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